

August 9, 2019

L-MT-19-039
10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

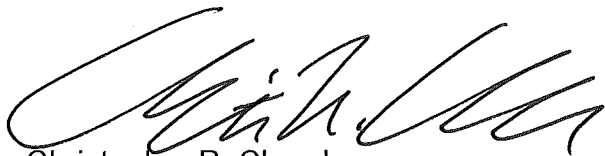
Monticello Nuclear Generating Plant
Docket No. 50-263
Renewed Facility Operating License No. DPR-22

LER 2019-002-00, Two Manual Primary Containment Isolation Valves Found Open Resulting in a Condition Prohibited by Technical Specification

Northern States Power Company, a Minnesota corporation, doing business as Xcel Energy hereby submits Monticello Nuclear Generating Plant (MNGP) Licensee Event Report (LER) 2019-002-00 "Two Manual Primary Containment Isolation Valves Found Open Resulting in a Condition Prohibited by Technical Specification" pursuant to 10 CFR 50.73(a)(2)(i)(B) as an operation or condition which was prohibited by the plant's Technical Specification; 10 CFR 50.73(a)(2)(v)(C), event or condition that could have prevented fulfillment of a safety function to control the release of radioactive material; and 10 CFR 50.73(a)(2)(v)(D), event or condition that could have prevented fulfillment of a safety function to mitigate the consequences of an accident.

Summary of Commitments

This letter makes no new commitments and no revisions to existing commitments.



Christopher R. Church
Site Vice President, Monticello Nuclear Generating Plant
Northern States Power Company – Minnesota

Enclosure

cc: Administrator, Region III, USNRC
Project Manager, Monticello Nuclear Generating Plant, USNRC
Resident Inspector, Monticello Nuclear Generating Plant, USNRC
Minnesota Department of Commerce



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name
Monticello Nuclear Generating Plant

2. Docket Number
05000-263

3. Page
1 OF 5

4. Title
Two Manual Primary Containment Isolation Valves Found Open Resulting in a Condition Prohibited by Technical Specification

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
06	11	2019	2019	002	00	08	09	2019	Facility Name	Docket Number
										05000
									Facility Name	Docket Number
										05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level 82%	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(iii)
	☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)		

12. Licensee Contact for this LER

Licensee Contact

Gus Hernandez

Telephone Number (Include Area Code)

(763)-271-6746

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To ICES	Cause	System	Component	Manufacturer	Reportable To ICES
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

15. Expected Submission Date

Month Day Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On 6/11/2019 during troubleshooting activities, two in-series manual 1" Primary Containment Isolation Valves (PCIVs), AS-78 and AS-79, were discovered to be open. In Mode 1 the required position of the PCIVs is closed. Investigation determined the valves were left open following a maintenance activity during a recent refueling outage. No formal tracking mechanism had been used for these valves to ensure plant configuration was restored following the maintenance activity as required by procedure. This event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) "as a condition was prohibited by the plant's Technical Specifications"; 10 CFR 50.73(a)(2)(v)(C), "event or condition that could have prevented fulfillment of a safety function to control the release of radioactive material"; and 10 CFR 50.73(a)(2)(v)(D), "event or condition that could have prevented fulfillment of a safety function to mitigate the consequences of an accident". There were no actual safety consequences from this condition. The immediate corrective action was to close the valves and independently verify they were locked closed. The applicable procedure will be revised to ensure clarity for when equipment manipulation per Operations Shift Supervisor direction is allowed and what formal tracking methods should be used to ensure equipment is returned to the required position.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Monticello Nuclear Generating Plant	05000-263	YEAR	SEQUENTIAL NUMBER	REV NO.
		2019	002	00

EVENT DESCRIPTION

On 6/11/2019 at approximately 11:32 while in Mode 1 at 82% power, a primary containment service air flowpath was found to be open. This service air flow path includes two in-series manual 1" Primary Containment Isolation Valves (PCIVs) [ISV]; AS-78, Drywell Isolation Valve Downstream and AS-79, Drywell Isolation Valve Upstream. These valves were found in the open position. In Mode 1 the required position of the PCIVs is closed. These valves are located in the steam chase, outside of Primary Containment (PCT) [BD], and are opened during refueling outages for maintenance activities.

The plant started Refueling Outage (RFO) 29 on 4/13/19. Near the completion of RFO 29, plant prestart checklists that included these valves were initiated on 5/7/19. The purpose of plant prestart checklists is to ensure that the plant is in a configuration to support startup and comply with plant operating license requirements.

During the Class 1 pressure test on 5/9/19, it was identified that three (3) Control Rod Drives (CRDs) were leaking under the reactor vessel due to worn o-ring seals.

The compressed air system prestart valve checklist verified the service air PCIVs, AS-78 and AS-79, were in a locked closed position at approximately 04:00 on 5/10/19.

During the subsequent 5/10/19 dayshift, 06:00 to 18:00, the decision was made to perform maintenance on the 3 leaking CRDs. Service air within PCT was needed to support the CRD repair. As part of the CRD repair work, AS-78 and AS-79 were re-opened.

The pre-job brief for this maintenance work did not meet management expectation to discuss plant configuration control requirements. There was no formal tracking mechanism of these open valves as required by procedure. The open status of these valves was also missed during shift turnover.

During the 5/11/19 dayshift, Operations completed the restoration from the CRD maintenance; however, because there was no formal tracking mechanism the service air PCIVs, AS-78 and AS-79, were not closed and remained open.

On 5/12/19 at 1858, the plant transitioned from Mode 4 to Mode 2 with the service air PCIVs open. PCIVs and PCT are required in Modes 1, 2, & 3.

PCT atmosphere was inerted with nitrogen to reduce the oxygen content to less than 4% by volume as required by the plant Technical Specifications on May 14, 2019 at 0307.

With these PCIVs open, plant service air leaked into the nitrogen filled PCT. On 5/25/19 at 1304, PCT oxygen analyzer control room alarm was received, and troubleshooting activities were initiated on 5/28/19. Troubleshooting actions included oxygen instrumentation calibration and leak testing, and verifying valve

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lineups.

On 6/11/2019 during a planned station down power to 82%, an Operator entered the steam chase as part of the investigation to validate the position of the service air PCIVs. The service air PCIVs were found open. These valves were then locked closed.

EVENT ANALYSIS

The PCIVs AS-78 and AS-79 were open and inoperable from 5/12/2019 until 06/11/2019. Once the valves were discovered open they were closed and independently verified locked closed at 1149 on 06/11/2019 (17 minutes after discovery). The event notification, 54111, was made on 06/11/2019 at 1655.

With both valves inoperable at the mode change from 4 to 2, Primary Containment limiting condition for operation (LCO) 3.6.1.1, Primary Containment Isolation Valves LCO 3.6.1.3, and LCO 3.0.4 were not met. With these valves open, analysis shows that Primary Containment Leakage Rate Testing Program TS program 5.5.11 requirements were not met. Also the Required Action for one or two inoperable PCIVs requires entry into Tech Spec LCO 3.6.1.3 Condition B which requires the station to isolate the affected penetration flow path by use of at least one closed and de-activated automatic valve, closed manual valve or blind flange within 1 hour was not met.

This condition was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) "as an operation or condition which was prohibited by the plant's Technical Specifications"; 10 CFR 50.73(a)(2)(v)(C), "event or condition that could have prevented fulfillment of a safety function to control the release of radioactive material"; and 10 CFR 50.73(a)(2)(v)(D), "event or condition that could have prevented fulfillment of a safety function to mitigate the consequences of an accident".

An engineering analysis determined that a safety system functional failure did not occur as defined in Nuclear Energy Institute (NEI) 99-02 Revision 7. PCT was able to perform its safety function and analysis showed that dose rates to station personnel and the public would not have exceeded regulatory dose limits.

SAFETY SIGNIFICANCE

There were no actual safety consequences from this condition. The function of the PCIVs, in combination with other accident mitigation systems, is to limit fission product release during and following postulated Design Basis Accidents (DBAs) to within limits. Primary containment isolation ensures that the release of radioactive material to the environment will be consistent with the assumptions used in the analyses for a DBA.

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An analysis was performed to determine the impact of AS-78 and AS-79 being open in accident analysis. The analysis determined that all other SSCs would perform their functions with AS-78 and AS-79 open. However, the leakage from PCT with the open valves would have exceeded the maximum allowable TS PCT leakage limits.

The analysis determined doses to the main control room (MCR), the exclusion area boundary (EAB), and the low population zone (LPZ). The analysis shows that with the assumed leakage rate, doses to the MCR, EAB, and LPZ were all within the regulatory limits.

Therefore, based on the aforementioned analysis results, the potential safety consequences of exceeding the maximum allowable TS primary containment leakage are very low due to radiation doses in the MCR, EAB, and LPZ all being within federal regulatory limits.

CAUSE

The direct cause of the condition was the inadequate adherence to the requirements of Operations Procedure to formally track the configuration of AS-78 and AS-79.

The root cause was that Operations leadership has accepted the inappropriate application of shift supervision authorization to change plant configuration.

Contributing cause 1: Inadequate application of management expectations to discuss configuration control questions.

Contributing cause 2: Turnover did not cover critical activities as described in site procedure.

CORRECTIVE ACTIONS

Upon discovery that the PCIVs were open, they were closed. The Operations procedure will be revised to:

- 1) allow equipment manipulation per Operations Shift Supervisor direction for emergency conditions only and
- 2) specify a tracking method to ensure equipment is returned to position.

Interim actions that were implemented:

- Shift Managers (SMs) attend all Operations Pre Job Briefs (PJBs).
- Plant Specific Configuration control questions will be asked at all PJB's.
- FP-OP-COO-17, Equipment Manipulation and Status Control, and FP-OP-SC-02, Alternate Plant

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Configuration Control, will be in hand and available for all PJB's.

- PJB's for planned evolutions SHALL be reverse brief.
- SM's will have a crew discussion to refresh knowledge on these standards.
- SM's will provide an EOS performance summary on PJB's to Operations Manager (OM), Assistant Operations Manager (AOM) and Shift Manager (SM population).
- Each crew (including Work Execution Center (WEC), Support and Fix It Now (FIN)), will draft a letter regarding commitment to configuration control and all members of the crew will sign the letter.
- Site-wide Memo was issued to communicate what happened, what went wrong, what went right and lessons to be learned and applied.

PREVIOUS SIMILAR EVENTS

There were no similar MNGP Licensee Event Reports during the past six years.

ADDITIONAL INFORMATION

The Institute of Electrical and Electronics Engineer codes for equipment are denoted by [XX].